

Hybrid-YOLO for classification of insulators defects in transmission lines based on UAV

Bruno José Souza ^{a,b,*}, Stefano Frizzo Stefenon ^{c,d}, Gurmail Singh ^e, Roberto Zanetti Freire ^a

^a Industrial and Systems Engineering Graduate Program (PPGEPS), Polytechnic School (EP), Pontifical Catholic University of Parana (PUCPR), Rua Imac. Conceição 1155, 80215-901 Prado Velho, Curitiba, Brazil

^b Pumatronix Electronic Equipment Ltd, Bartolomeu L. de Gusmão 1970, 81650-050 Curitiba, Brazil

^c Digital Industry Center, Fondazione Bruno Kessler, Via Sommarive 18, 38123 Povo, Trento, Italy

^d Department of Mathematics, Computer Science, and Physics, University of Udine, Via delle Scienze 206, 33100 Udine, Italy

^e Department of Computer Sciences, University of Wisconsin-Madison, 1210 W. Dayton Street, Madison, WI 53706-1613, United States

ARTICLE INFO

Keywords:

Deep learning
Power grid inspection
You only look once

ABSTRACT

Transmission power lines are essential to supply electrical energy to consumption centers. Keeping a reliable transmission system requires the early identification of faults. Image-based inspection of transmission lines makes fault identification faster and more accessible since it can be carried out using unmanned aerial vehicles (UAVs) in hard-to-reach places. In this paper, it is proposed to use a hybrid version of the You Only Look Once (YOLO) using ResNet-18 classifier, for power system inspection based on real images of failed components recorded by UAVs. This work assumed a dataset including 1,593 power grid inspection pictures for a supervised training. Based on YOLOv5x, with an mAP of 0.99262, the proposed method was superior to YOLOv5n, YOLOv5s, YOLOv5m, and YOLOv5l for object detection. For the multiclassification task, with an F1_score result of 0.96216, the proposed Hybrid-YOLO was superior to distinct architectures as the VGG-11, VGG-13, VGG-16, VGG-19, ResNet-34, ResNet-50, ResNet-152, DenseNet-121, DenseNet-161, DenseNet-169, DenseNet-201, YOLOv5, YOLOv6, and YOLOv7 versions.

1. Introduction

The transmission lines are responsible for taking electricity from the generation source to the large consumption centers. Typically these systems are installed outdoors, in regions with difficult access [1]. To guarantee the supply of electric energy, inspections are carried out in the transmission system to identify failures before shutdowns [2]. Failures can range from contamination that accumulates in the network to missing insulators, which are a serious problem for network isolation [3].

The contamination that accumulates on the surface of transmission line insulators increases their surface conductivity, making the location more susceptible to electrical discharges [4]. When an insulator has its surface charred, it loses its insulating capacity, making the system vulnerable. Broken insulators are a common fault found on transmission lines, as their surface is often made of glass that can break due to depredation. When an insulator is broken, it must be replaced because it represents a problem for the transmission system that can lead to a network power failure [5].

The identification of failures during inspection can be done using specific equipment such as ultrasonic detectors [6], radio interference [7], and other acoustic methods [8]. The use of this equipment to detect partial discharges requires operators with high experience, electrical machines can generate interference in measurements, which makes it a difficult task to classify an adverse condition [9]. The biggest difficulty in using sound-based methods to identify faults is that the measurement needs to be close to the network, i.e., the inspection team needs to pass by the branch line and take measurements even in places that are difficult to access [10].

Performing the analysis of the electrical system through imaging is gaining space and can be done even at a considerable distance [11]. Ultraviolet and infrared images can be used to facilitate the identification of defects in the power electrical system [12]. Based on this need, unmanned aerial vehicles (UAVs) are increasingly used to carry out aerial inspections through imaging, mainly in places of difficult access [13].

The great advantage of using UAVs is the ability to analyze a branch on which there are terrain variations that make it difficult for ground

* Corresponding author.

E-mail address: bruno.jose@pucpr.edu.br (B.J. Souza).

inspection teams to access [14]. For this reason, inspections using UAVs have been analyzed by several researchers in the last few years as in [15,16] where object detection techniques are used, and in [17,18] where semantic segmentation and binocular stereo vision were applied. Through advanced image classification techniques using deep learning, it is possible to classify adverse conditions quickly and accurately, making this approach promising to improve the ability to carry out inspections of the electrical power system [19].

Using UAVs to perform inspections in the electrical power system, Zhai et al. [20] proposed an algorithm based on spatial morphological characteristics. Based on this approach it is possible to identify faults in glass and ceramic insulators. From the morphological process, the fault is highlighted with high accuracy and can be applied to improve inspections. According to Wang et al. [21], a promising way to identify faults in insulators is based on infrared images. Based on this approach, it would be possible to use embedded systems for predictive maintenance.

The model that has been outstanding for object detection, being superior to well-established models is the You Only Look Once (YOLO) [22], which is also applied for real-time object detection systems [23]. The Hybrid-YOLO has been improved and is more efficient, the proposed method outperforms previous versions in terms of mean average precision and F1_score. For this reason, in this paper, the analysis using this model was evaluated for a dataset recorded by UAVs in an inspection of the electrical power grid.

The main contributions of this paper are:

- The Hybrid-YOLO proves to be superior for insulator detection to other models using a small dataset, achieving better results in evaluation metrics than YOLOv5n, YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. The use of small databases is a challenge in computer vision since there is a lack of information to train the model effectively.
- The proposed hybrid method based on the ResNet-18 classifier proved to be superior for the multiclassification task, being possible to identify different conditions by changing the classifier, achieving better F1_score results than standard models. Identifying faulty insulators during electrical power system inspections is a difficult task since there is a similarity between good and faulty components.
- The proposed approach has a low computational effort, so this model has a higher potential to be adapted for embedded systems and can be used in online applications using UAVs. This advantage is intrinsic to the method since YOLO identifies objects in a single step, making it faster than classic slide window-based models.

The remainder of this paper is organized as follows: Section 2 presents background about the use of UAVs in electrical power grid inspections. In Section 3 the proposed method is addressed and the considerations for the analysis of this paper. Section 4 presents the results of the application of the method regarding the variation in its parameters. Finally, in Section 5 a conclusion is drawn.

2. Power grid inspection using unmanned aerial vehicles

Insulator strings that have missing insulators are commonly observed in electrical inspections. Because this type of failure occurs often in transmission lines [24], several authors carry out an analysis to identify this condition and determine the place where corrective maintenance should be performed. Fahim et al. [25] developed an analysis related to the identification of the insulators using the single-shot multibox detector. Identifying the position of insulators is necessary when the objective is to locate missing components in the insulator chain.

According to Liu et al. [26], the improvement in specific algorithms for detecting objects is something that shows promise when inspections

are carried out on transmission lines, as standard models have difficulty classifying images with diverse background interference. To perform the inspection and identify components that have faults in the transmission line, it is common to use the UAV, as it is versatile and accessible equipment nowadays [27].

The detection of multiple types of faults makes the inspection more suitable to be performed by the electric power utility company using UAVs since besides missing insulators other faults can be found in the power system [28]. Contamination on structures caused by bird nests can be easily identified in images recorded by UAVs, resulting in an F1_score of up to 96.87% considering a faster region-based convolutional neural network (R-CNN) [29].

From the use of the UAV, the inspection can be performed autonomously based on convolutional neural networks (CNN), in a simple way, since commercial drones have the necessary characteristics to be used for this application [30]. An issue that deserves attention when using autonomous UAVs is the gradual accumulation of errors according to the drone's movement. To minimize the impact of error measurement, the good estimation of the equipment position improves its robustness in this analysis, resulting in an efficiency of 91.44% in field applications [31].

The high intensity of electric fields near transmission lines is a factor that also needs to be considered when using UAVs. A strong electric field affects UAV sensors and can lead to aircraft instability. To reduce the impact of this interference filters are applied to correct the interference from the sensors and guarantee the stability of the aerial inspection [32]. Using a finite element model Chen et al. [33] evaluate the safety distance to use the UAV for transmission line inspections. The electromagnetic compatibility limit should be analyzed to have a safe power line inspection.

To use UAVs in the inspection of the power grid, the optimized trajectory needs to be defined. The intensity of the magnetic field is a parameter that can be used to have a safe trajectory during inspections [34]. According to Liu, Wang, and Liu [35], one of the difficulties in performing long trajectory inspections occurs due to the flight limitation of UAVs due to their battery capacity. Trajectory automation makes inspection more efficient, based on route planning the UAVs with automatic multi-rotor can improve the inspection by up to 62.88% and can be applied to the large-scale inspection of the power grid [36].

The issue of the cumulative error generated by updating the positioning of UAVs needs to be taken into consideration during inspections of the transmission lines. Liu et al. [37] propose the use of the least square polynomial curve fitting method as the correlation scan may mismatch under the geometric feature condition. The positioning of UAVs can be improved through strategies that separate the towers into system nodes, facilitating inspection over long distances [38].

To perform an analysis efficiently feature of extraction techniques to improve model performance, Miao et al. [25] proposed a single-shot multibox detector (SSD) to detect insulators in aerial images. This approach can realize automatic feature extraction and improve the speed of identifying the insulators from images with complex backgrounds. In addition to finding where are the insulators, these deep learning structures are applied to identify foreign objects that may result in faults in the power grid [39].

The combination of techniques often shows promising results in terms of accuracy regarding the classification of adverse conditions in transmission lines [40]. This is because there are models that are specific for identifying objects, and models that are appropriate for classifying adverse conditions [41]. Thus, the combination of a classifier model and an object identification model can result in a significant improvement in the ability to classify adverse conditions in transmission lines, such as the identification of defective insulators [42].

The use of aerial images to identify faults is becoming more and more explored, as it is an indirect analysis, in which it is not necessary to have contact with the network, specific training is not necessary

Table 1
Related works.

Author	Number of images	Main method	Data augmentation	Identif. of insul. chain	Fault location
Sampedro et al. [51]	160	CNN	✓	✓	✓
Jiang et al. [52]	385	SSD	✓	✓	✓
Zhang et al. [46]	400	SWT	✓	×	✓
Tao et al. [30]	600	CNN	✓	✓	✓
Wen et al. [49]	879	Exact R-CNN	✓	✓	✓
Li et al. [44]	1600	Faster R-CNN	✓	✓	✓
Zhao et al. [24]	1843	AVSCNet	×	×	✓
Wang et al. [40]	2560	ResNeSt	✓	✓	✓
Li et al. [29]	2700	Faster R-CNN	✓	×	✓
Han et al. [28]	4031	CNN	✓	✓	✓
Liu et al. [26]	5000	YOLOv3	✓	✓	×
Liu et al. [42]	5847	RetinaNet	×	✓	✓
Miao et al. [25]	6700	SSD	✓	✓	×
Wang et al. [45]	6864	SCA	✓	×	✓
Zhu et al. [39]	8000	SHM	✓	×	✓
Zhao et al. [53]	10,468	Faster R-CNN	×	✓	✓

since UAVs are increasingly popular [43], in addition to being able to be a comprehensive analysis as it is possible to train the model for the task that is desired based on the problem of the location. Segmentation and feature extraction techniques become promising alternatives to improve the model's ability to handle a large volume of images with distinct backgrounds [44].

According to Wang et al. [45], UAVs generate many raw images and are an alternative to carrying out the classification of defects through image segmentation. Using line accessory detection, background filter, and artificial contour segment, more representative images are obtained for the classification of failures. To highlight the fault, the use of adaptive threshold segmentation associated with gray variance normalization proves to be a promising alternative for applications involving the analysis of power grid degradation [46].

Identifying faults in insulators is difficult due to the position of the observer or the camera since in some situations it is not possible to view the damaged part of the component. Field inspection teams usually view insulators from the ground, and may have difficulty identifying faults that are on the top of the insulator, because they have a bottom-up view [47]. Seeking to improve the detection performance of inclined objects, Wang et al. [48] proposed the R-YOLO model, which rotates the bounding box during the Intersection over Union computation. Considering that the images have variations in the capture angle due to the route of the UAV, adding inspection images to the training helps the model to achieve better performance.

Using deep learning models it becomes possible to identify defects in transmission lines based on images captured by UAVs [49]. According to Kim et al. [50], the use of cameras with higher light spectrum capacity increases the capacity to identify faults even in high-resolution images, and is an alternative to improve inspections. Considering that some authors performed the analysis regarding the location of the fault, others only focused on the identification of the position of the insulator. Table 1 presents a comparative analysis of what was performed in this field of study. As the small number of images is a challenge for training the network, in this table, the number of images in the dataset and the strategy assumed for the analysis are also presented.

As can be observed, many authors use data augmentation strategies, since the number of failures recorded during inspections can be small. When the objective is to locate faults in insulator chains, the database is usually less than 5000 images, only in Li et al. [29] faults in insulators were identified using a larger dataset. However, other types of faults are analyzed, then a larger dataset is considered. Zhao et al. [53] applied a different approach using 4 datasets, they used 2×3234 and 2×2000 , totaling 10,468 images.

Some of the authors develop studies on the positioning of UAVs, so they are not presented in this comparison. Besides the well-established methods for object detection, other methods based on computer vision

are used in inspections such as automatic visual shape clustering network (AVSCNet) [24], scale histogram matching (SHM) [39], segment connection algorithm (SCA) [45], and square wave transformation (SWT) [46].

The total number of images, as well as the number of images per class, in the computer vision tasks solved by machine learning, are important because directly influence the results obtained through training, considering that with more images the model has a higher learning capacity. In the YOLOv5 repository [54], it is suggested in its hints section that approximately 1500 images per class are indicated for the appropriate use of this model. In the next section, information about the structure of the YOLO model is presented.

YOLOv5 is increasingly being applied, Zhu et al. [55] proposed the use of this model with transformer prediction heads (TPH), thereby applying the TPH-YOLOv5 for object detection based on UAVs scenarios. Liu et al. [56] presented an improved YOLOv5 based on transfer learning for UAVs application. Focusing on power generation system fault detection, Zhang et al. [57] employed YOLOv5 for the inspection of photovoltaic modules using UAVs. Shen, Lin, and Song [58] proposed a method for object detection deployed on UAVs that outperforms the standard YOLOv5.

For power electrical system inspection, Peng et al. [59] had a 1.32% improvement in the mAP@[0.5] in its Res-YOLOv5 version compared to the original YOLOv5. Especially applied for fault detection, Feng et al. [60] used the YOLOv5x for defects detection of insulators, achieving an mAP of 95.5%. Li et al. [61] had an mAP of 97.82% for the same goal. Since contamination is a common problem in power grids, Ge et al. [62] used the YOLOv5 for the identification of strange objects on transmission lines.

The improvement of the YOLOv5 model is promising [63], as presented by Gao, Chen, and Lin [64], and Zhang et al. [65] for insulator detection. There are several variations of YOLOv5 that can be employed for electrical system fault detection [66], Feng and Jiang [67] compared the YOLOv5x to YOLOv5s regarding the recognition of insulators. Huang, Hu and Zhang [68] have proposed improvements in the YOLOv5s for identifying faults in insulators of electrical power grids. Based on YOLOv5s Zhang et al. [69] proposed the Prune-MobileNetv3-YOLOv5s for moving object recognition of the distribution power grid.

As presented in this paper, the fault classification task can be separated from the goal of insulator detection. Wang et al. [70] applied the YOLOv5s for the insulator detection from transmission lines. Zhang et al. [71] used the YOLOv5s to improve the safety of workers in the construction of the distribution network. Cheng and Lu [72] have evaluated an improved YOLOv5s for protective gear detection, focused on the maintenance of high-voltage equipment.

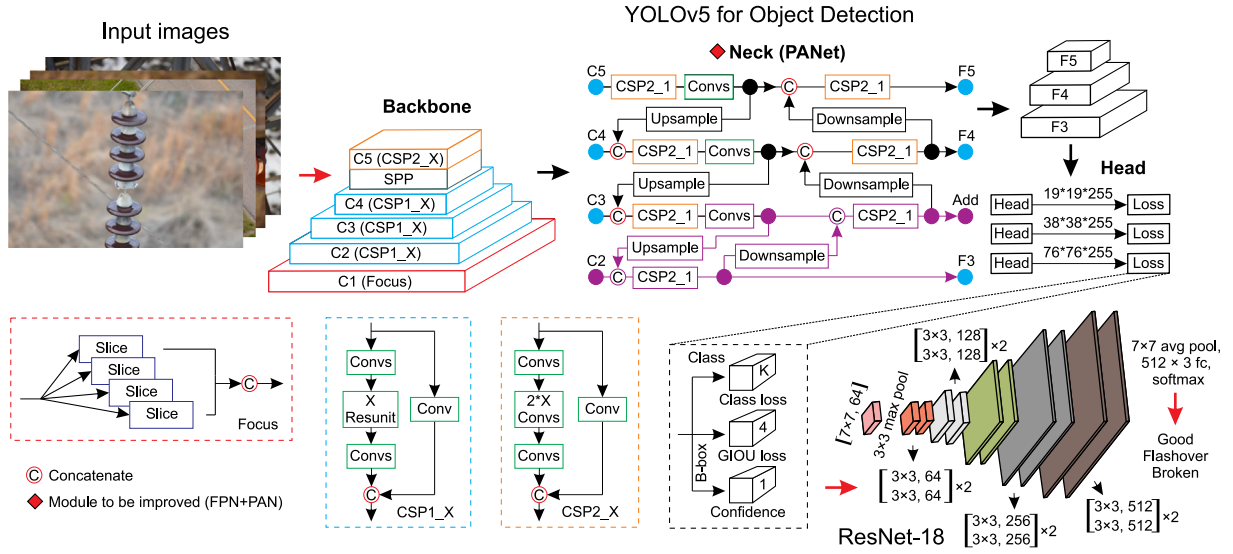


Fig. 1. Structure of the proposed method.

3. Proposed method

The YOLO algorithm is a convolutional neural network that divides images into grids, with each cell in the grid detecting objects within itself. This algorithm is one of the single-shot detectors, and the network gets its name from the fact that it only needs to process the image once to classify or detect the object in question [73]. In recent years, the YOLO models have demonstrated good results in object detection, and the model has undergone several improvements since its initial versions [74].

In this paper, the YOLOv5 is used for object detection, and a ResNet-18 is applied for the classification task. The structure of the proposed method is presented in Fig. 1, where the PANet improvement in the YOLOv5 over the neck is highlighted in purple. Each step of the proposed method is evaluated to determine the best approach for each specific task.

3.1. YOLOv5 structure

The input images are divided into $S \times S$ grid, each grid space is responsible for predicting the target bounding box centered, corresponding to its degree of confidence. Each grid cell predicts probability classes $pr(class_i|object)$, then the confidence is given by:

$$pr(class_i|object) \cdot pr(object) \cdot IOU_{pred}^{truth} = pr(class_i) \cdot IOU_{pred}^{truth} \quad (1)$$

when the grid contains predicted objects $pr(object)$ equals to 1, otherwise it is 0. After dividing the image into grids a class probability map is created to identify the desired objects and bounding boxes to determine whether the desired objects are located in this confidence region.

The YOLOv5 structure has characteristics that make it promising for object detection, and consequently for fault detection in insulators, which is the objective of this paper. The model uses Mosaic to improve the detection performance of small objects. Through Mosaic, an increase in the number of data is introduced in the training process, which indirectly increases the batch size, and makes each iteration receive more data. This reduces the target size and therefore increases the detection efficiency of smaller objects [75].

This variation of YOLO uses the techniques Focus, BottleneckCSP, SPP, and PANet to improve the object detection capability of the model. The Focus is used to enhance the receptive field, BottleneckCSP extracts the feature information, SPP separates the relevant features from the context to improve the non-linear representation capacity of the network, and PANet integrates the high and low-level characteristics to

Table 2

YOLOv5 models setup information for 640-pixel images.

Model	Depth multiple	Width multiple	Layers	Params (M)	FLOPs
YOLOv5n	0.33	0.25	213	1.8	4.1
YOLOv5s	0.33	0.50	213	7.0	15.8
YOLOv5m	0.67	0.75	290	20.9	47.9
YOLOv5l	1.00	1.00	367	46.1	107.7
YOLOv5x	1.33	1.25	444	86.2	203.8

improve the detection accuracy and achieve a better stronger feature map [76].

The YOLOv5 versions, released by Ultralytics [54], were pre-trained on the common objects in context (COCO) dataset. COCO is a large-scale dataset that can be used for object detection, having 330k images, 1.5 million object instances, and 80 object categories [77]. The versions that use 640-pixel images are YOLOv5n (nano), YOLOv5s (small), YOLOv5m (medium), YOLOv5l (large), and YOLOv5x (extra large). The difference between these versions is the size of the model, given by the number of layers and parameters used, these size variations are generated by depth multiple and width multiple. The number of parameters and floating-point operations (FLOPs) is different from the model presented by Ultralytics [54] since the used dataset has a different number of images and classes.

To obtain better results, it is conventional to scale up a baseline detector by using larger backbone networks. The paper presented by Tan, Pang, and Le [78] presents an alternative idea named compound scaling (EfficientDet method), which is used in YOLOv5 models. The architecture of the EfficientDet uses a weighted bi-directional feature pyramid network (BiFPN) to multi-scale feature fusion and a compound scaling technique that uniformly scales the resolution, therefore, it jointly scales up all dimensions of the backbone.

The Depth multiple parameters are in charge of the depth of the model i.e. it ultimately adds more layers to the neural network, while the Width Multiple is responsible to add more filters into the layers so it adds more channels to the layer outputs. These configurations' variations result in the differences between the YOLOv5n, YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x versions. The relationship between Depth Multiple and Width Multiple and the size of each model is presented in Table 2.

Compared to other versions of YOLOv5 the YOLOv5x proves to be more accurate, with an $mAP_{0.5}^{val}$ of 68.9 using images of 640 pixels [54], analyzing the COCO dataset. The YOLOv5x has higher FLOPs, this does

not represent a problem for this work considering that the analysis is performed offline. In the YOLO versions, it is noticeable the influence of the number of parameters on the model efficiency [60], it is observed that the use of fewer parameters results in higher model speed but, generally, lower mAP [79].

Since YOLO models are very promising for object detection, this paper proposes the use of a CNN classifier, applying the YOLO for insulator detection. Structures based on Visual Geometry Group (VGG), Residual Neural Network (ResNet), and Dense Convolutional Network (DenseNet) architectures were considered. Particularly, the VGG-11, VGG-13, VGG-16, VGG-19 [80], ResNet-18, ResNet-34, ResNet-50, ResNet-152 [81], DenseNet-121, DenseNet-161, DenseNet-169, and DenseNet-201 [82] were evaluated.

3.2. Application of the previous version

According to Dewi et al. [74], the YOLOv4 reaches an accuracy of up to 89.33%, while YOLOv3 reaches 84.9% when an application based on the recognition of traffic signs is considered. Comparatively, even earlier versions of YOLO, such as YOLOv3 prove superior to well-established algorithms used for object detection [23]. The YOLOv3 proved to be faster than RetinaNet [83], Faster R-CNN [84], feature pyramid network with Faster R-CNN [85], and DSSD513 [86], among other models. In the analysis presented by Farhadi and Redmon [23], the YOLOv3 was faster and presents better mAP results than other models.

Sadykova et al. [87] applied YOLO for UAV-based electrical system inspection. According to his analysis, it is necessary to consider data augmentation to avoid overfitting when the dataset is large, which is the case with inspection images of transmission lines considering high-voltage insulators. Some difficulties in classifying failures must be considered, such as the presence of light variations, different backgrounds, and a large dataset [88].

Specifically, to detect insulator failures during electrical power system inspections, Liu et al. [26] used the modified YOLOv3 model achieving an average precision of 94.47% for failure identification. The standard YOLOv3 and YOLOv2 result in an accuracy of 90.31% and 83.43%, respectively, which shows that changing the model can be a good strategy.

Introducing a modified version of YOLO denoted modified YOLO-tiny for insulator (MTI-YOLO), Liu et al. [89] achieved an average precision 17% higher than YOLO-tiny and 9% higher than YOLO-v2 for insulator detection using UAV images. The running time of the MTI-YOLO was slightly higher, with a lower memory usage of 25.6% and 38.9% than YOLO-v2 and YOLO-v3, respectively, on images with diverse backgrounds interference.

The use of YOLO is becoming increasingly popular for fault detection of insulators in power transmission lines [60]. This model has stood out for this task, being applied by Chen et al. [90], Liu et al. [91], and Chen et al. [92]. With the evolution in the YOLO model and the superior performance results, the YOLOv5 present in this paper shows promise for evaluating inspection images recorded by UAVs.

3.3. Dataset

The dataset used in this paper is available at [93], being high-resolution photographs of insulator chains with defective components. The insulators in this database are porcelain in various colors, registered in diverse positions with different image backgrounds, as can be seen in Fig. 2. This dataset has 3 sub-classes: flashover damage insulator shell, broken insulator shell, and good insulator shell. The dataset has a total of 1596 images, of which 1116 are used for training, 160 for validation, and 320 for testing. The components are labeled based on bounding boxes regarding their vertical and horizontal position, their width, and their height.

3.4. Experiment

The experiment is separated into 2 analyses, regarding the detection and classification of the insulators. To perform the training of the YOLOv5 versions for detecting the insulators, 1016 images were used for training, and 179 were used for validation from the dataset [93]. For the classification 12,827 cutouts were used, wherein 10,886 were for training and 1941 for validation. The same number of images was used to evaluate the model from the results of the preprocessing techniques.

The use of a batch size greater than 16 makes the process slower, without a significant improvement in the results, for this reason, a batch size limit equal to 16 was used, this value is the one suggested for the YOLOv5 versions by Ultralytics [54]. The algorithms evaluated in this paper were developed using Python language. To perform the training of the models, the following requirements were allocated for each experiment: 64 GB of RAM (random access memory) and an NVIDIA GeForce GTX 1080 Ti GPU (Graphics Processing Unit). To perform the statistical analysis, 8 GPUs were used in parallel with these same requirements. The training was performed by adjusting the image resolution to 640×640 .

There are several ways to evaluate the performance of a computer vision algorithm, where the metrics are derived from information obtained from the confusion matrix. In this paper, the precision, recall, and F1_score were evaluated, these measures are given by:

$$\text{precision} = \frac{tp}{tp + fp}, \quad (2)$$

$$\text{recall} = \frac{tp}{tp + fn}, \quad (3)$$

$$\text{F1_score} = \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}}, \quad (4)$$

where tp is the true positive, fn is the false negative, and fp is the false positive.

Especially for object detection tasks, the intersection over union (IoU) is a criterion that determines if the detected object and the real object are similar. It is defined by the intersection area divided by the union of the object area and the detected area [94]. A tp detection is defined by $\text{IoU} > T$, where T is a predefined threshold. In this work, the evaluation is made for T equal to 0.5. The average precision (AP) summarizes a precision–recall curve as the weighted mean of precision achieved at each threshold is calculated according to:

$$\text{AP} = \sum_{\eta} (\text{recall}_{\eta} - \text{recall}_{\eta-1}) \text{precision}_{\eta}. \quad (5)$$

where η is the n th threshold. The mean average precision (mAP) is calculated by taking the mean AP over all classes, given by:

$$\text{mAP} = \frac{1}{n} \sum_{k=1}^n \text{AP}_k, \quad (6)$$

wherein n is the number of classes and k is the corresponding class.

Employing the model that obtained the best result for object detection, the statistical analysis was performed to evaluate the stability in the model performance regarding the $\text{mAP}@[0.5]$, considering that the F1_score is used to evaluate the classification results and the mAP for the evaluation of object detection. To achieve this, 50 simulations are conducted from a random global training seed. The values of minimum, maximum, mean, median, standard deviation, and variance of the $\text{mAP}@[0.5]$ are presented. Since there is the re-sampling in each simulation, the quantity $\text{mAP}@[0.5]$ is now a random variable with an unknown mean (μ). A threshold value of 0.99 was defined, considering that in the preliminary analysis the 3 best models had results close to this value, this being an acceptable limit value for the analysis in question. Afterward, the following hypothesis test is evaluated:

Is the true mean of the $\text{mAP}@[0.5]$ of the best model equal to or higher than 0.99?

$$\begin{cases} H_0 : \mu \geq 0.99 \\ H_1 : \mu < 0.99. \end{cases} \quad (7)$$



Fig. 2. Images of insulator chains from the dataset.

A hypothesis test places side by side two hypotheses about the population that gave rise to the available data sample. To solve this problem, there is the null hypothesis (H_0) and the alternative hypothesis (H_1). In this paper, a left-tailed test is used, in which the null hypothesis is that the value is greater than or equal to the specified value. Based on this test the p -value is obtained, the p -value is a measure of the probability that an observed difference could have occurred just by random chance [95].

Additionally, the Shapiro–Wilk normality test is evaluated, this test checks the null hypothesis that a sample $x_1, \dots, x_n$ came from a normally distributed population. In the Shapiro–Wilk test, the null hypothesis supports that the variable is normally distributed, and the alternative hypothesis states that the variable is not normally distributed [96]. At the end of the evaluation of the proposed model, a benchmark is presented using standard YOLOv5 [54], YOLOv6 [97], and YOLOv7 [98] versions.

4. Results and discussion

Selecting the appropriate model to achieve the best result in fault identification is a difficult task since there are many variations of well-established models. The big difference in the choice of model is based on the available data set, considering that in some cases using networks with many layers makes the analysis inefficient. There is a trade-off between the appropriate model, concerning available resources, database, and the need for greater speed to be computed.

The goal of this paper is to achieve the best object detection and classification result, disregarding the time needed for training since the training is performed offline. For this reason, the definition of the optimal structures and the fine-tuning are conducted regarding the performance results of the mAP and the F1_score metrics. For the object detection task, the focus will be on finding the best mAP, and for the classification task finding the best F1_score.

For a proper comparison among these models, it is necessary to set a maximum number of epochs, in a preliminary analysis, the model that needed more epochs to converge was YOLOv5x, requiring 780 epochs until there was an early stopping where there is no further improvement in training (after 50 epochs) and the model completes the training avoiding overfitting. The model that converged the fastest was YOLOv5n, requiring 546 epochs, the loss functions for this model are presented in Fig. 3.

Considering that none of the versions of the YOLOv5 needed more than 1000 epochs to converge, this was set as the maximum value of epochs, while observing that early stopping was maintained when

overfitting occurs, thus ensuring proper training. Therefore, a comparative analysis between the YOLOv5 versions for the object detection task was performed. In this section, the best results of each model are highlighted in bold, and the overall best is underlined.

4.1. Structure definition for object detection

Considering the specific object detection task, Table 3 presents the results regarding the use of the YOLOv5n, YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x models, varying the batch size. The larger models took longer to converge, which is expected considering that they use more parameters. As the objective of this paper is to find the model with the best classification results, the computational effort was not considered a limiting factor, considering the resources used for the experiments.

It is observed that as the batch size is larger the models tend to converge faster, on the other hand, they need a GPU with a larger memory to load the data. For this reason, there is a trade-off between objective processing time and available hardware. To enable the reproducibility of the experiment, a maximum batch size of 16 was set.

The initial analysis of the YOLOv5 versions shows that the YOLOv5x had the best result considering the mAP@[0.5] equal to 0.99262. Since this metric is defined to evaluate object detection, which is the focus of this step, the YOLOv5x model using a batch size equal to 8 was set as the default model for object detection. In general, models using fewer parameters (YOLOv5n, YOLOv5s, and YOLOv5m) had lower results (under 0.99 in some cases) regarding mAP@[0.5].

The differences between the standard YOLOv5 versions are related to the parameters “depth multiple” and “width multiple”, which result in a progressive increase in the model. The progressive improvement in the results according to the size of the model matches with the results of the comparisons made with other databases, such as the COCO dataset for instance [54].

Comparing the results of the same model using smaller batch sizes the variations obtained were small. This information is important since using larger models requires more GPU memory and using a larger batch size also leads to this need. This means that it is possible to compute larger models with smaller batch sizes, without overloading GPU memory, while maintaining satisfactory results.

The YOLOv5n model did not converge faster using a batch size of 16 than using a batch size equal to 8, which was expected. This result occurred because in this simulation the model converged in 546 epochs using a batch size of 8, whereas it converged after 887 epochs using a batch size of 16. This shows that although the batch size has a great influence on the computational effort, the random initialization of the

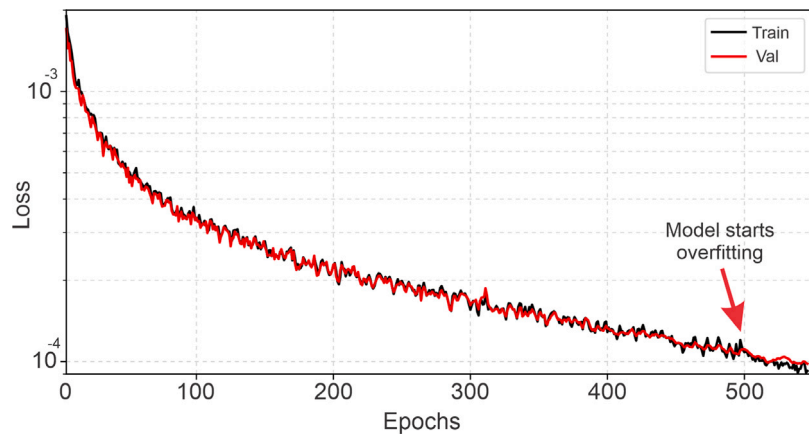


Fig. 3. Training vs. validation loss.

Table 3
Object detection evaluation of YOLOv5 models.

Model	Batch size	Precision	Recall	F1_score	mAP		Mem. (GB)	Time (h)
					[0.5]	[0.5:0.95]		
YOLOv5n	2	0.97454	0.95775	0.96607	0.98189	0.84057	0.24	12.57
	4	0.98171	0.96806	0.97484	0.98638	0.86059	0.54	6.48
	8	0.98148	0.97063	0.97602	0.98585	0.85870	1.00	5.78
	16	0.98394	0.97854	0.98123	0.98888	0.89250	1.90	7.93
YOLOv5s	2	0.98294	0.97966	0.98129	0.98622	0.88866	0.61	20.93
	4	0.98643	0.98506	0.98574	0.98944	0.91939	0.88	12.60
	8	0.98761	0.98566	0.98663	0.98986	0.91680	1.78	10.45
	16	0.98506	0.98490	0.98498	0.99070	0.93208	3.44	8.41
YOLOv5m	2	0.98558	0.98605	0.98581	0.98984	0.92390	1.11	26.36
	4	0.98861	0.98406	0.98633	0.99078	0.94457	1.89	13.23
	8	0.98765	0.98854	0.98809	0.99255	0.94759	3.19	11.79
	16	0.98953	0.98918	0.98935	0.99177	0.95390	5.93	11.30
YOLOv5l	2	0.98209	0.98851	0.98529	0.99065	0.94735	1.94	29.09
	4	0.98613	0.98866	0.98739	0.99171	0.95268	2.97	21.37
	8	0.98683	0.98764	0.98723	0.99039	0.95398	5.18	15.73
	16	0.98868	0.98962	0.98915	0.99050	0.95867	9.03	13.98
YOLOv5x	2	0.98538	0.98506	0.98522	0.99005	0.93945	3.21	30.91
	4	0.98547	0.98918	0.98732	0.99152	0.95744	4.63	39.08
	8	0.98346	0.98970	0.98657	0.99262	0.95742	7.70	23.15
	16	0.98987	0.98557	0.98771	0.99130	0.95517	10.9	16.33

model can make a significant difference in the results of processing time. Following, a complete evaluation of the YOLOv5x model for object detection is presented.

4.2. Detailed assessment using YOLOv5x

This section aims to present the results obtained during the experiment using YOLOv5x for insulator detection. Fig. 4 presents the results related to the training of the model. Fig. 4 was divided into four subfigures where each one presents the evaluation of a distinct metric during the epochs in the training phase. Since the model converges at 780 epochs, the results are presented up to this point.

In sub Fig. 4(a) (precision) is verified that for the validation data, the model achieves results above 95% from 20 training epochs, reaching 98.92% at its best, this shows that the model is fast to achieve satisfactory results during training. The results of recall values are presented in sub Fig. 4(b), this evaluated metric represents the total number of correct detection in percentage. The recall value reached a total of 98.98% during the training.

The sub Figs. 4(c) and 4(d) are responsible for presenting the results of mean average precision 0.50 and 0.50:0.95 respectively. The notation 0.50:0.95 means average mAP over different IoU thresholds, from 0.5 to 0.95. As in the other subfigures, the performance of the model presents a similar dynamic, increasing considerably after the

20th epoch and with greater stability after half of the training. The best value for mAP@[0.5] was 99.35%, while for mAP@[0.5:0.95] it was 95.79%.

Considering that YOLOv5x was superior for insulators detection, regarding mAP@[0.5], a statistical evaluation is presented in Table 4 for this model. The goal of the statistical analysis is to assess variability in the results as a metric of the reliability of the model. To accomplish the statistical analysis, 50 experiments with different initialization seeds were executed using the same hardware requirements to speed up the evaluation. In this case, 8 GPUs were used in parallel on a cluster, as explained in Section 3.4.

There is not a considerable difference between the minimum and maximum values in the statistical analyses regarding the mAP@[0.5], just this 2.88% in 50 runs. Consequently, the standard deviation and variance values are low, 6.64×10^{-4} and 4.41×10^{-7} respectively, showing that the model has little variability and is therefore a reliable model for the analysis presented. The mean and median values remain acceptable for the use of this model.

In the hypothesis test, with a mean of 0.9919 and a standard deviation of 6.64×10^{-4} , the p -value of 1.0 was obtained from a level of confidence of 5%, which indicates non-rejection of H_0 . That is, there is no reason to reject the hypothesis that the mAP is greater than or equal to 0.99. This confirms that there is a high probability of getting an acceptable mAP. In the Shapiro-Wilk normality test, the p -value

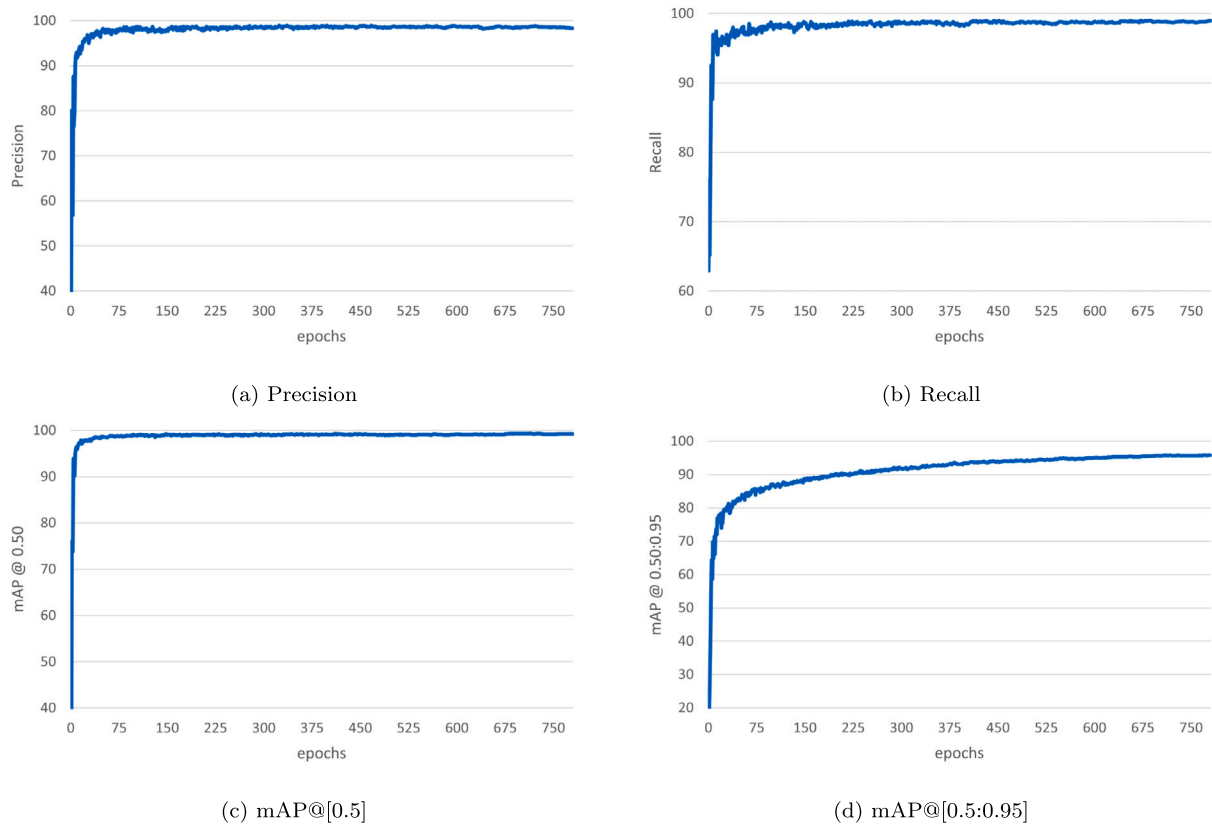


Fig. 4. Metrics comparison during training.

Table 4
Statistical evaluation of the YOLOv5x model.

Measure	Min	Max	Mean	Median	Std. Dev.	Variance
Precision	0.98602	0.99224	0.98940	0.98921	1.31×10^{-3}	1.72×10^{-6}
Recall	0.98300	0.99073	0.98769	0.98764	1.44×10^{-3}	2.07×10^{-6}
F1_score	0.98577	0.99039	0.98854	0.98840	9.24×10^{-4}	8.53×10^{-7}
mAP@[0.5]	0.99083	0.99369	0.99190	0.99180	6.64×10^{-4}	4.41×10^{-7}
mAP@[0.5:0.95]	0.94629	0.96303	0.95853	0.95959	3.82×10^{-3}	1.46×10^{-5}

of 0.1174 is obtained. Based on this result, the null hypothesis can be acceptable. Thus, there is evidence that the data tested is normally distributed. This assumption is based on the null hypothesis of this test, which is to have a normally distributed population.

To demonstrate the operation of the model, an analysis of YOLOv5x is performed using the test dataset, regarding insulator detection. This database has a total of 320 images as specified in Section 3.3. When evaluating the model in the test phase, the results were 0.9217 for the precision, 0.9908 for the recall, and 0.9550 for the F1_score. Comparing the results obtained in the test dataset, it is seen that the precision value ends up dropping from 0.9835 to 0.9217, this occurred by the presence of some false positive detection made by the model. An example of this false positive detection can be seen in Fig. 5, where the model ends up finding an insulator where there is none. This presentation ends up being polluted due to the overlapping of the objects. Despite the lower precision results in the test phase, with an F1_score of 0.9550, the model is still acceptable in the insulator detection task.

Fig. 6 aims to present assertive detection performed by the YOLOv5x in the test dataset. It is observed that the model can have difficulties because the objects are mostly small. In addition, due to the angle of the images, the insulators are often superimposed, and therefore the bounding boxes used in the training can contain information from more than one instance, which can cause decreases in the model's performance.

4.3. Classifier

Considering that several models can be used to perform insulator classification, Table 5 presents a comparison between well-established CNN-based models. The VGG-based models provided the worst results when compared to the other CNN structures, with the best result of F1_score of 0.89238 obtained by the VGG-19 model. In terms of processing time, there was a trend toward more time needed for the models to converge depending on the number of model parameters. The DenseNet-161 model had an F1_score of 0.95384, showing better results for the classification task than all VGG structures.

The best classifier considering the F1_score was ResNet-18 obtaining a result of 0.96216, for this reason, this model was used as the default classifier in this paper. The processing time was not considered to determine which model was more appropriate for the classification, however, it can be observed that ResNet-18 was faster. This result shows that for some challenges the use of larger models may not result in the best results. Using ResNet-18 as the base classifier, benchmarking against the standard YOLO versions is performed.

4.4. Benchmarking

To validate the proposed model, in Table 6 a comparison to standard YOLO versions, considering the multiclassification task is presented.

Table 5
CNN classifier evaluation.

Model	Precision	Recall	F1_score	Time (h)
VGG-11	0.86224	0.81249	0.83663	51.83
VGG-13	0.91758	0.81862	0.86528	76.08
VGG-16	0.93854	0.78873	0.85714	89.67
VGG-19	0.94444	0.84577	0.89238	94.56
ResNet-18	0.99441	0.93193	0.96216	21.92
ResNet-34	0.98351	0.93717	0.95978	34.05
ResNet-50	0.96739	0.86829	0.91516	78.03
ResNet-152	0.96756	0.88177	0.92268	173.33
DenseNet-121	0.97849	0.89215	0.93333	99.28
DenseNet-161	0.99465	0.91625	0.95384	108.67
DenseNet-169	0.97872	0.89756	0.93638	117.46
DenseNet-201	0.97395	0.93969	0.95652	128.08

Table 6
Benchmarking of YOLO models.

Model	Precision	Recall	F1_score	mAP	
				[0.5]	[0.5:0.95]
YOLOv5n	0.89221	0.81341	0.85099	0.89165	0.68488
YOLOv5s	0.91744	0.89828	0.90776	0.94012	0.76101
YOLOv5m	0.94856	0.91806	0.93306	0.95606	0.81775
YOLOv5l	0.95489	0.93070	0.94264	0.96168	0.84154
YOLOv5x	0.96346	0.92702	0.94489	0.95698	0.84900
YOLOv6n	0.93370	0.91160	0.92031	0.94765	0.80871
YOLOv6t	0.93563	0.91518	0.92399	0.94972	0.80852
YOLOv6s	0.94756	0.90572	0.92411	0.95125	0.81645
YOLOv6m	0.93231	0.91898	0.92394	0.95706	0.82841
YOLOv6l	0.94870	0.93618	0.94131	0.95892	0.85859
YOLOv7 (std)	0.90815	0.91423	0.91118	0.96005	0.76091
YOLOv7x	0.91893	0.92227	0.92060	0.95944	0.76341
Hybrid-YOLO	0.99441	0.93193	0.96216	0.99262	0.95742

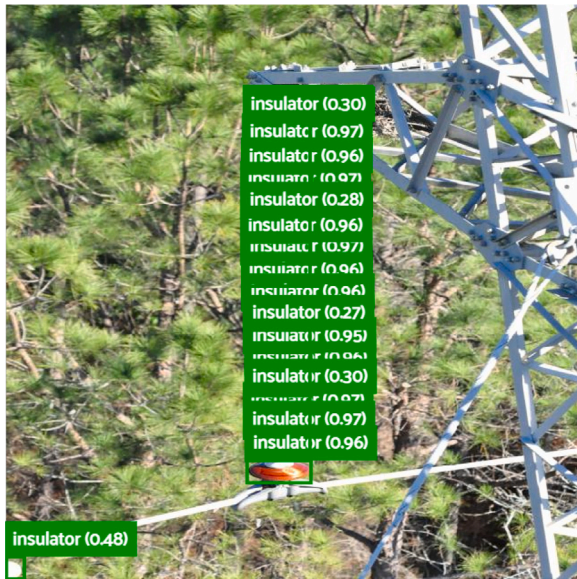


Fig. 5. Example of false positive detection.

The first observation about the results is that YOLOv5x was superior to the other standard models for fault classification, this shows that this model is promising for both, insulator identification and classification.

It was noticed that there is a tendency to obtain better results of F1_score in the models that use more parameters. The best overall results were achieved by the proposed model in all metrics evaluated, showing that the proposed methodology is promising. When a specific CNN classifier is used, there is an improvement in the F1_score

results, moreover, the YOLOv5x base model is shown to be more precise when only one insulator is considered than in multiclassification, been this model is suitable specifically for insulator detection with an mAP@[0.5] of 0.99262.

The difference between the versions of the YOLOv6 (n, t, s, m, and l) [97] and YOLOv7 (standard and x) [98] are also based on the Depth Multiple and Width Multiple, as is the version of YOLOv5 [54]. Regarding model evaluation, YOLOv6 presented by Li et al. [97] needs to be executed in two steps to evaluate the model, with the training results being separated from the evaluation results, which makes it more difficult to analyze the model, and also this model has less flexibility in the configuration settings for training. The YOLOv7 presented by Wang, Bochkovskiy, and Liao [98] is a similar version to the YOLOv5 released by Ultralytics [54].

5. Final remarks

The use of UAVs is increasingly popular due to the technological advances in aircraft control and the low cost of small equipment. The inspection of the electrical system becomes considerably faster when this equipment is used, since the electrical power grid, in some cases, is installed in places of difficult access. By using the proposed method with images captured from UAVs, the inspection of the power grid can be performed more easily and at a lower cost, increasing the reliability of the electric power system.

The Hybrid-YOLO has shown promise for the identification of various conditions of transmission insulators. The combination of the best model for insulator detection with the best CNN structure for insulator condition classification provided a hybrid approach that outperforms standard models. Based on ResNet-18 the proposed method reaches the F1_score of 0.96216 and the mAP@[0.5] of 0.99262. When compared to other variations of YOLO such as YOLOv5 (n, s, m, l, and x), YOLOv6 (n, t, s, m, and l), and YOLOv7 (std and x) the proposed approach outperformed all the variations included in the comparison.

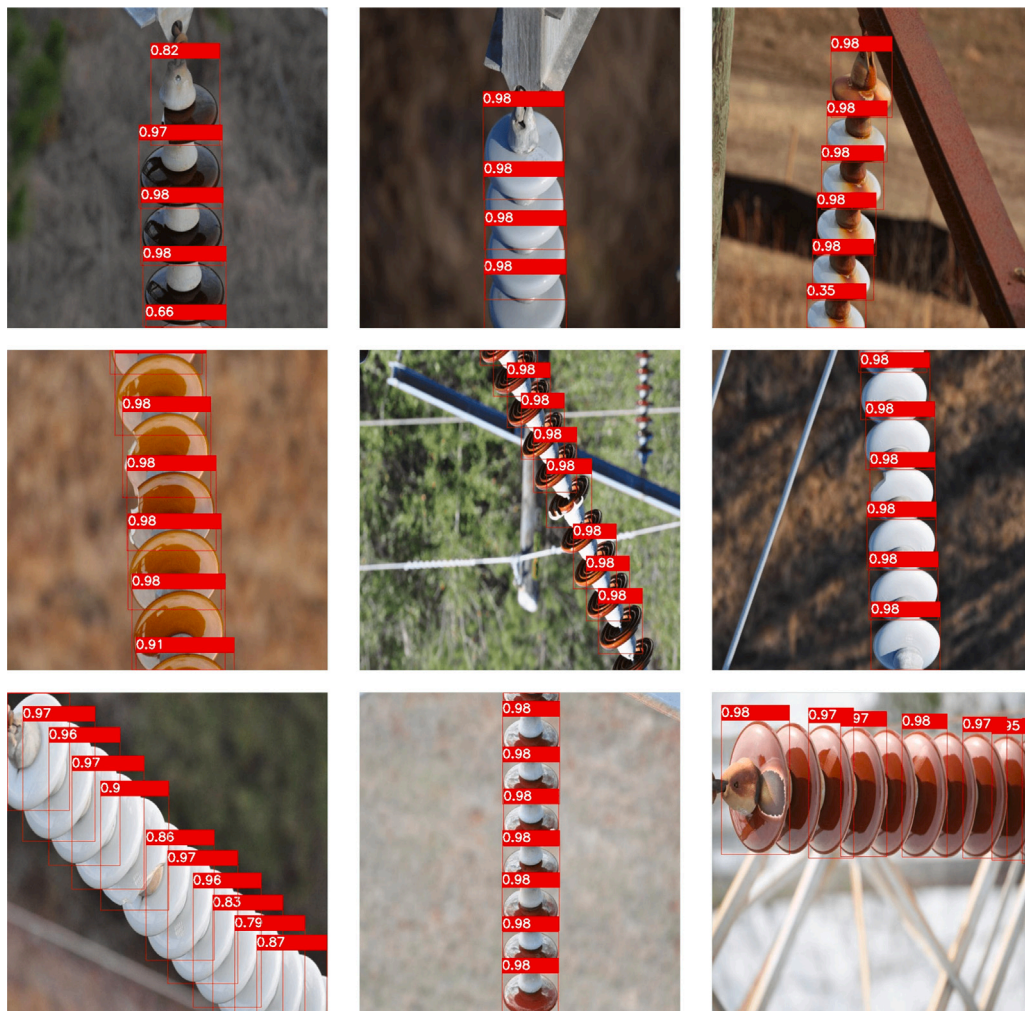


Fig. 6. Test results insulator detection.

The models with more parameters had promising results and the batch size did not result in a significant loss in the evaluated metrics, reducing the batch size can be an alternative to computing the models when there is a hardware limitation. Specifically, in the YOLOv5x model, the reduction in batch size from 16 to 8 resulted in a difference of 0.1155% (lower) in the F1_score and 0.1332% (higher) in the mAP@[0.5] for the object detection task.

Future work can be done focusing on the implementation of the Hybrid-YOLO, presented in this paper, in an embedded system. With an embedded system that has global positioning system tracking it is possible to define a routine to inspect the system and notify the maintenance team of the locations where signs of failure are identified automatically, increasing the speed of identification of faults and maintenance of the electrical power grid.

CRediT authorship contribution statement

Bruno José Souza: Computational analysis. **Stefano Frizzo Stefenon:** Wrote about the problem, Applied methods, Evaluated the results. **Gurmail Singh:** Performed the deep CNN analysis, Final overall review of the work. **Roberto Zanetti Freire:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors appreciate the scholarship provided by Pumatronix Electronic Equipment Ltd for the development of this research.

References

- [1] Corso MP, Stefenon SF, Couto VF, Cabral SHL, Nied A. Evaluation of methods for electric field calculation in transmission lines. *IEEE Lat Am Trans* 2018;16(12):2970–6. <http://dx.doi.org/10.1109/TLA.2018.8804264>.
- [2] El-Hag A. Application of machine learning in outdoor insulators condition monitoring and diagnostics. *IEEE Instrum Meas Mag* 2021;24(2):101–8. <http://dx.doi.org/10.1109/MIM.2021.9400959>.
- [3] Stefenon SF, Freire RZ, Meyer LH, Corso MP, Sartori A, Nied A, Klaar ACR, Yow K-C. Fault detection in insulators based on ultrasonic signal processing using a hybrid deep learning technique. *IET Sci, Meas Technol* 2021;14(10):953–61. <http://dx.doi.org/10.1049/iet-smt.2020.0083>.
- [4] Stefenon SF, Singh G, Yow K-C, Cimatti A. Semi-ProtoPNet deep neural network for the classification of defective power grid distribution structures. *Sensors* 2022;22(13):4859. <http://dx.doi.org/10.3390/s22134859>.
- [5] Stefenon SF, Neto CSF, Coelho TS, Nied A, Yamaguchi CK, Yow K-C. Particle swarm optimization for design of insulators of distribution power system based on finite element method. *Electr Eng* 2022;104:615–22. <http://dx.doi.org/10.1007/s00202-021-01332-3>.
- [6] Yao Z, Yu X, Yao J, Sui W, Yu X. Transmission line insulator fault detection based on ultrasonic technology. In: *Journal of physics: Conference series*, vol. 1187. IOP Publishing; 2019, 022056.

- [7] Anjum S. A study of the detection of defects in ceramic insulators based on radio frequency signatures (Master's thesis), University of Waterloo; 2014.
- [8] Park K-C, Motai Y, Yoon JR. Acoustic fault detection technique for high-power insulators. *IEEE Trans Ind Electron* 2017;64(12):9699–708. <http://dx.doi.org/10.1109/TIE.2017.2716862>.
- [9] Polisetty S, El-Hag A, Jayram S. Classification of common discharges in outdoor insulation using acoustic signals and artificial neural network. *High Volt* 2019;4(4):333–8. <http://dx.doi.org/10.1049/hve.2019.0113>.
- [10] Stefenon SF, Seman LO, Sopelsa Neto NF, Meyer LH, Nied A, Yow K-C. Echo state network applied for classification of medium voltage insulators. *Int J Electr Power Energy Syst* 2022;134:107336. <http://dx.doi.org/10.1016/j.ijepes.2021.107336>.
- [11] Ibrahim A, Dalbah A, Abualsaud A, Tariq U, El-Hag A. Application of machine learning to evaluate insulator surface erosion. *IEEE Trans Instrum Meas* 2020;69(2):314–6. <http://dx.doi.org/10.1109/TIM.2019.2956300>.
- [12] Jin L, Tian Z, Ai J, Zhang Y, Gao K. Condition evaluation of the contaminated insulators by visible light images assisted with infrared information. *IEEE Trans Instrum Meas* 2018;67(6):1349–58. <http://dx.doi.org/10.1109/TIM.2018.2794938>.
- [13] Waleed D, Mukhopadhyay S, Tariq U, El-Hag AH. Drone-based ceramic insulators condition monitoring. *IEEE Trans Instrum Meas* 2021;70:1–12. <http://dx.doi.org/10.1109/TIM.2021.3078538>.
- [14] Nguyen VN, Janssen R, Roverso D. Intelligent monitoring and inspection of power line components powered by UAVs and deep learning. *IEEE Power Energy Technol Syst J* 2019;6(1):11–21. <http://dx.doi.org/10.1109/JPETS.2018.2881429>.
- [15] Han J, Yang Z, Xu H, Hu G, Zhang C, Li H, Lai S, Zeng H. Search like an eagle: A cascaded model for insulator missing faults detection in aerial images. *Energies* 2020;13(3):713. <http://dx.doi.org/10.3390/en13030713>.
- [16] Rahman EU, Zhang Y, Ahmad S, Ahmad HI, Jobaer S. Autonomous vision-based primary distribution systems porcelain insulators inspection using UAVs. *Sensors* 2021;21(3):974. <http://dx.doi.org/10.3390/s21030974>.
- [17] Wang L, Chen Z, Hua D, Zheng Z. Semantic segmentation of transmission lines and their accessories based on UAV-taken images. *IEEE Access* 2019;7:80829–39. <http://dx.doi.org/10.1109/ACCESS.2019.2923024>.
- [18] Ma Y, Li Q, Chu L, Zhou Y, Xu C. Real-time detection and spatial localization of insulators for UAV inspection based on binocular stereo vision. *Remote Sens* 2021;13(2):230. <http://dx.doi.org/10.3390/rs13020230>.
- [19] Siddiqui ZA, Park U. A drone based transmission line components inspection system with deep learning technique. *Energies* 2020;13(13):3348. <http://dx.doi.org/10.3390/en13133348>.
- [20] Zhai Y, Chen R, Yang Q, Li X, Zhao Z. Insulator fault detection based on spatial morphological features of aerial images. *IEEE Access* 2018;6:35316–26. <http://dx.doi.org/10.1109/ACCESS.2018.2846293>.
- [21] Wang B, Dong M, Ren M, Wu Z, Guo C, Zhuang T, Pischler O, Xie J. Automatic fault diagnosis of infrared insulator images based on image instance segmentation and temperature analysis. *IEEE Trans Instrum Meas* 2020;69(8):5345–55. <http://dx.doi.org/10.1109/TIM.2020.2965635>.
- [22] Jocher G, Stoken A, Borovec J, NanoCode012, Chaurasia A, TaoXie, Changyu L, V A, Laughing, tkianai, yxNONG, Hogan A, lorenzomamma, AlexWang1900, Hajek J, Diaconu L, Marc, Kwon Y, oleg, wanghaoyang0106, Defretin Y, Lohia A, mlSah, Milanko B, Fineran B, Khromov D, Yiwei D, Doug, Durgesh, Ingham F. ultralytics/yolov5: v5.0 - YOLOv5-P6 1280 models, AWS, Supervise.ly and YouTube integrations. 2021. <http://dx.doi.org/10.5281/zenodo.4679653>.
- [23] Farhadi A, Redmon J. Yolov3: An incremental improvement. In: *Computer vision and pattern recognition*. Berlin, Germany: Springer; 2018, p. 1804–02767.
- [24] Zhao Z, Qi H, Qi Y, Zhang K, Zhai Y, Zhao W. Detection method based on automatic visual shape clustering for pin-missing defect in transmission lines. *IEEE Trans Instrum Meas* 2020;69(9):6080–91. <http://dx.doi.org/10.1109/TIM.2020.2969057>.
- [25] Miao X, Liu X, Chen J, Zhuang S, Fan J, Jiang H. Insulator detection in aerial images for transmission line inspection using single shot multibox detector. *IEEE Access* 2019;7:9945–56. <http://dx.doi.org/10.1109/ACCESS.2019.2891123>.
- [26] Liu C, Wu Y, Liu J, Sun Z. Improved YOLOv3 network for insulator detection in aerial images with diverse background interference. *Electronics* 2021;10(7):771. <http://dx.doi.org/10.3390/electronics10070771>.
- [27] Hui X, Bian J, Zhao X, Tan M. Vision-based autonomous navigation approach for unmanned aerial vehicle transmission-line inspection. *Int J Adv Robot Syst* 2018;15(1):1–15. <http://dx.doi.org/10.1177/1729881417752821>.
- [28] Han J, Yang Z, Zhang Q, Chen C, Li H, Lai S, Hu G, Xu C, Xu H, Wang D, Chen R. A method of insulator faults detection in aerial images for high-voltage transmission lines inspection. *Appl Sci* 2019;9(10):2009. <http://dx.doi.org/10.3390/app9102009>.
- [29] Li F, Xin J, Chen T, Xin L, Wei Z, Li Y, Zhang Y, Jin H, Tu Y, Zhou X, Liao H. An automatic detection method of bird's nest on transmission line tower based on faster-RCNN. *IEEE Access* 2020;8:164214–21. <http://dx.doi.org/10.1109/ACCESS.2020.3022419>.
- [30] Tao X, Zhang D, Wang Z, Liu X, Zhang H, Xu D. Detection of power line insulator defects using aerial images analyzed with convolutional neural networks. *IEEE Trans Syst, Man, Cybern: Syst* 2020;50(4):1486–98. <http://dx.doi.org/10.1109/TSMC.2018.2871750>.
- [31] Menéndez O, Pérez M, Auat Cheein F. Visual-based positioning of aerial maintenance platforms on overhead transmission lines. *Appl Sci* 2019;9(1):165. <http://dx.doi.org/10.3390/app9010165>.
- [32] da Silva MF, Honório LM, Marcato ALM, Vidal VF, Santos MF. Unmanned aerial vehicle for transmission line inspection using an extended Kalman filter with colored electromagnetic interference. *ISA Trans* 2020;100:322–33. <http://dx.doi.org/10.1016/j.isatra.2019.11.007>.
- [33] Chen D-Q, Guo X-H, Huang P, Li F-H. Safety distance analysis of 500kV transmission line tower UAV patrol inspection. *IEEE Lett Electromagn Compat Pract Appl* 2020;2(4):124–8. <http://dx.doi.org/10.1109/LEMCPA.2020.3040878>.
- [34] Wu Y, Zhao G, Hu J, Ouyang Y, Wang SX, He J, Gao F, Wang S. Overhead transmission line parameter reconstruction for UAV inspection based on tunneling magnetoresistive sensors and inverse models. *IEEE Trans Power Deliv* 2019;34(3):819–27. <http://dx.doi.org/10.1109/TPWRD.2019.2891119>.
- [35] Liu Z, Wang X, Liu Y. Application of unmanned aerial vehicle hangar in transmission tower inspection considering the risk probabilities of steel towers. *IEEE Access* 2019;7:159048–57. <http://dx.doi.org/10.1109/ACCESS.2019.2950682>.
- [36] He T, Zeng Y, Hu Z. Research of multi-rotor UAVs detailed autonomous inspection technology of transmission lines based on route planning. *IEEE Access* 2019;7:114955–65. <http://dx.doi.org/10.1109/ACCESS.2019.2935551>.
- [37] Liu C, Shao Y, Cai Z, Li Y. Unmanned aerial vehicle positioning algorithm based on the secant slope characteristics of transmission lines. *IEEE Access* 2020;8:43229–42. <http://dx.doi.org/10.1109/ACCESS.2020.2977923>.
- [38] Jiang C, Xie J, Zhao J, Zhang X. A novel high-voltage transmission line joint temperature monitoring system using hybrid communication networks. *IEEE Access* 2021;9:109478–87. <http://dx.doi.org/10.1109/ACCESS.2021.3097372>.
- [39] Zhu J, Guo Y, Yue F, Yuan H, Yang A, Wang X, Rong M. A deep learning method to detect foreign objects for inspecting power transmission lines. *IEEE Access* 2020;8:94065–75. <http://dx.doi.org/10.1109/ACCESS.2020.2995608>.
- [40] Wang S, Liu Y, Qing Y, Wang C, Lan T, Yao R. Detection of insulator defects with improved ResNeSt and region proposal network. *IEEE Access* 2020;8:184841–50. <http://dx.doi.org/10.1109/ACCESS.2020.3029857>.
- [41] Zhang P, Zhang Z, Hao Y, Zhou Z, Luo B, Wang T. Multi-scale feature enhanced domain adaptive object detection for power transmission line inspection. *IEEE Access* 2020;8:182105–16. <http://dx.doi.org/10.1109/ACCESS.2020.3027850>.
- [42] Liu J, Jia R, Li W, Ma F, Abdullah HM, Ma H, Mohamed MA. High precision detection algorithm based on improved RetinaNet for defect recognition of transmission lines. *Energy Rep* 2020;6:2430–40. <http://dx.doi.org/10.1016/j.egy.2020.09.002>.
- [43] Kim S, Kim D, Jeong S, Ham J-W, Lee J-K, Oh K-Y. Fault diagnosis of power transmission lines using a UAV-mounted smart inspection system. *IEEE Access* 2020;8:149999–50009. <http://dx.doi.org/10.1109/ACCESS.2020.3016213>.
- [44] Li X, Su H, Liu G. Insulator defect recognition based on global detection and local segmentation. *IEEE Access* 2020;8:59934–46. <http://dx.doi.org/10.1109/ACCESS.2020.2982288>.
- [45] Wang L, Chen Z, Hua D, Zheng Z. Semantic segmentation of transmission lines and their accessories based on UAV-taken images. *IEEE Access* 2019;7:80829–39. <http://dx.doi.org/10.1109/ACCESS.2019.2923024>.
- [46] Zhang Y, Huang X, Jia J, Liu X. A recognition technology of transmission lines conductor break and surface damage based on aerial image. *IEEE Access* 2019;7:59022–36. <http://dx.doi.org/10.1109/ACCESS.2019.2914766>.
- [47] Corso MP, Perez FL, Stefenon SF, Yow K-C, García Ovejero R, Leithardt VRQ. Classification of contaminated insulators using k-nearest neighbors based on computer vision. *Computers* 2021;10(9):112. <http://dx.doi.org/10.3390/computers10090112>.
- [48] Wang X, Zheng S, Zhang C, Li R, Gui L. R-YOLO: A real-time text detector for natural scenes with arbitrary rotation. *Sensors* 2021;21(3):888. <http://dx.doi.org/10.3390/s21030888>.
- [49] Wen Q, Luo Z, Chen R, Yang Y, Li G. Deep learning approaches on defect detection in high resolution aerial images of insulators. *Sensors* 2021;21(4):1033. <http://dx.doi.org/10.3390/s21041033>.
- [50] Kim S, Kim D, Jeong S, Ham J-W, Lee J-K, Oh K-Y. Fault diagnosis of power transmission lines using a UAV-mounted smart inspection system. *IEEE Access* 2020;8:149999–50009. <http://dx.doi.org/10.1109/ACCESS.2020.3016213>.
- [51] Sampedro C, Rodríguez-Vázquez J, Rodríguez-Ramos A, Carrio A, Campoy P. Deep learning-based system for automatic recognition and diagnosis of electrical insulator strings. *IEEE Access* 2019;7:101283–308. <http://dx.doi.org/10.1109/ACCESS.2019.2931144>.
- [52] Jiang H, Qiu X, Chen J, Liu X, Miao X, Zhuang S. Insulator fault detection in aerial images based on ensemble learning with multi-level perception. *IEEE Access* 2019;7:61797–810. <http://dx.doi.org/10.1109/ACCESS.2019.2915985>.
- [53] Zhao W, Xu M, Cheng X, Zhao Z. An insulator in transmission lines recognition and fault detection model based on improved faster RCNN. *IEEE Trans Instrum Meas* 2021;70:1–8. <http://dx.doi.org/10.1109/TIM.2021.3112227>.
- [54] Ultralytics G. YOLOv5 in PyTorch. 2022. <https://github.com/ultralytics/yolov5>.
- [55] Zhu X, Lyu S, Wang X, Zhao Q. TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios. In: 2021 IEEE/CVF international conference on computer vision workshops (ICCVW). Montreal, Canada: IEEE; 2021, p. 2778–88. <http://dx.doi.org/10.1109/ICCVW54120.2021.00312>.

- [56] Liu W, Quijano K, Crawford MM. YOLOv5-tassel: Detecting tassels in RGB UAV imagery with improved YOLOv5 based on transfer learning. *IEEE J Sel Top Appl Earth Obs Remote Sens* 2022;15:8085–94. <http://dx.doi.org/10.1109/JSTARS.2022.3206399>.
- [57] Zhang X, Zou P, Ma C, Zhang Z, Guo H, Chen Y, Cheng Z. Inspection and classification system of photovoltaic module defects based on UAV and thermal imaging. In: 2022 7th International conference on power and renewable energy (ICPRE). Shanghai, China: IEEE; 2022, p. 905–9. <http://dx.doi.org/10.1109/ICPRE55555.2022.9960506>.
- [58] Shen H, Lin D, Song T. Object detection deployed on UAVs for oblique images by fusing IMU information. *IEEE Geosci Remote Sens Lett* 2022;19:1–5. <http://dx.doi.org/10.1109/LGRS.2022.3141109>.
- [59] Peng G, Lei Y, Li H, Wu D, Wang J, Liu F. CORY-Net: Contrastive res-YOLOv5 network for intelligent safety monitoring on power grid construction sites. *IEEE Access* 2021;9:160461–70. <http://dx.doi.org/10.1109/ACCESS.2021.3132301>.
- [60] Feng Z, Guo L, Huang D, Li R. Electrical insulator defects detection method based on YOLOv5. In: 2021 IEEE 10th data driven control and learning systems conference (DDCLS). Suzhou, China: IEEE; 2021, p. 979–84. <http://dx.doi.org/10.1109/DDCLS52934.2021.9455519>.
- [61] Li Q, Zhao F, Xu Z, Wang J, Liu K, Qin L. Insulator and damage detection and location based on YOLOv5. In: 2022 International conference on power energy systems and applications (ICOPESA). Singapore: IEEE; 2022, p. 17–24. <http://dx.doi.org/10.1109/ICOPESA54515.2022.9754476>.
- [62] Ge Z, Li H, Yang R, Liu H, Pei S, Jia Z, Ma Z. Bird's nest detection algorithm for transmission lines based on deep learning. In: 2022 3rd International conference on computer vision, image and deep learning & international conference on computer engineering and applications (CVIDL & ICEA). Changchun, China: IEEE; 2022, p. 417–20. <http://dx.doi.org/10.1109/CVIDLICEA56201.2022.9824057>.
- [63] Liu M, Li Z, Li Y, Liu Y. A fast and accurate method of power line intelligent inspection based on edge computing. *IEEE Trans Instrum Meas* 2022;71:1–12. <http://dx.doi.org/10.1109/TIM.2022.3152855>.
- [64] Gao J, Chen X, Lin D. Insulator defect detection based on improved YOLOv5. In: 2021 5th Asian conference on artificial intelligence technology (ACAIT). Haikou, China: IEEE; 2021, p. 53–8. <http://dx.doi.org/10.1109/ACAIT53529.2021.9731270>.
- [65] Zhang Z-D, Zhang B, Lan Z-C, Liu H-C, Li D-Y, Pei L, Yu W-X. FiNet: An insulator dataset and detection benchmark based on synthetic fog and improved YOLOv5. *IEEE Trans Instrum Meas* 2022;71:1–8. <http://dx.doi.org/10.1109/TIM.2022.3194909>.
- [66] Shi C, Lin L, Sun J, Su W, Yang H, Wang Y. A lightweight YOLOv5 transmission line defect detection method based on coordinate attention. In: 2022 IEEE 6th Information technology and mechatronics engineering conference (ITOEC). 6, Chongqing, China: IEEE; 2022, p. 1779–85. <http://dx.doi.org/10.1109/ITOEC53115.2022.9734540>.
- [67] Feng H, Jiang Y. Recognition of insulator based on yolov5 algorithm. In: 2021 IEEE 11th annual international conference on cyber technology in automation, control, and intelligent systems (CYBER). Jiaxing, China: IEEE; 2021, p. 505–9. <http://dx.doi.org/10.1109/CYBER53097.2021.9588319>.
- [68] Huang Z, Hu S, Zhang L. Fault Detection of insulator in distribution network Based on YOLOv5s Neural Network. In: 2022 International conference on artificial intelligence and computer information technology (AICIT). Yichang, China: IEEE; 2022, p. 1–5. <http://dx.doi.org/10.1109/AICIT55386.2022.9930315>.
- [69] Zhang Y, Yuan B, Zhang J, Li Z, Pang C, Dong C. Lightweight PM-YOLO network model for moving object recognition on the distribution network side. In: 2022 2nd Asia-Pacific conference on communications technology and computer science (ACCTCS). Shenyang, China: IEEE; 2022, p. 508–16. <http://dx.doi.org/10.1109/ACCTCS53867.2022.00109>.
- [70] Wang J, Zhang T, Xue X, Chen L. Real-time recognition of transmission line insulators under complex backgrounds: A YOLOv5s approach. In: 2022 4th International conference on power and energy technology (ICPET). Beijing, China: IEEE; 2022, p. 77–83. <http://dx.doi.org/10.1109/ICPET55165.2022.9918361>.
- [71] Zhang Y, Shi J, Wang D, Pang C, Yang Z, Wu B, Xu Y, Du W. Detection on safety helmet wearing of distribution network construction based on YOLOv5-btri algorithm. In: 2022 2nd Asia-Pacific conference on communications technology and computer science (ACCTCS). Shenyang, China: IEEE; 2022, p. 517–24. <http://dx.doi.org/10.1109/ACCTCS53867.2022.00110>.
- [72] Cheng X, Lu T. An improved YOLOv5s for protective gear detection. In: 2022 7th International conference on intelligent computing and signal processing (ICSP). Xi'an, China: IEEE; 2022, p. 661–5. <http://dx.doi.org/10.1109/ICSP54964.2022.9778822>.
- [73] Redmon J, Divvala S, Girshick R, Farhadi A. You only look once: Unified, real-time object detection. In: 2016 IEEE Conference on computer vision and pattern recognition (CVPR). Las Vegas, USA: IEEE; 2016, p. 779–88. <http://dx.doi.org/10.1109/CVPR.2016.91>.
- [74] Dewi C, Chen R-C, Liu Y-T, Jiang X, Hartomo KD. Yolo V4 for advanced traffic sign recognition with synthetic training data generated by various GAN. *IEEE Access* 2021;9:97228–42. <http://dx.doi.org/10.1109/ACCESS.2021.3094201>.
- [75] Lin F, Hou T, Jin Q, You A. Improved YOLO based detection algorithm for floating debris in waterway. *Entropy* 2021;23(9):1111. <http://dx.doi.org/10.3390/e23091111>.
- [76] Wang H, Zhang S, Zhao S, Wang Q, Li D, Zhao R. Real-time detection and tracking of fish abnormal behavior based on improved YOLOV5 and SiamRPN++. *Comput Electron Agric* 2021;106512. <http://dx.doi.org/10.1016/j.compag.2021.106512>.
- [77] Lin T-Y, Maire M, Belongie S, Hays J, Perona P, Ramanan D, Dollár P, Zitnick CL. Microsoft COCO: Common objects in context. In: European conference on computer vision – ECCV 2014, Cham, Switzerland: Springer; 2014, p. 740–55. http://dx.doi.org/10.1007/978-3-319-10602-1_48.
- [78] Tan M, Pang R, Le QV. Efficientdet: Scalable and efficient object detection. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020, p. 10781–90.
- [79] Zhao J, Ma J, Xin J, An R. Application of YOLOv5 algorithm in identification of transmission line insulators. In: International conference on mobile multimedia communications. Mobile multimedia communications, Cham, Switzerland: Springer; 2021, p. 880–7. http://dx.doi.org/10.1007/978-3-030-89814-4_65.
- [80] Nair RR, Singh T, Basavapattana A, Pawar MM. Multi-layer, multi-modal medical image intelligent fusion. *Multimedia Tools Appl* 2022;1–27. <http://dx.doi.org/10.1007/s11042-022-13482-y>.
- [81] Zhang L, Li H, Zhu R, Du P. An infrared and visible image fusion algorithm based on ResNet-152. *Multimedia Tools Appl* 2022;81(7):9277–87. <http://dx.doi.org/10.1007/s11042-021-11549-w>.
- [82] Lu T, Han B, Chen L, Yu F, Xue C. A generic intelligent tomato classification system for practical applications using DenseNet-201 with transfer learning. *Sci Rep* 2021;11(1):1–8. <http://dx.doi.org/10.1038/s41598-021-95218-w>.
- [83] Lin T-Y, Goyal P, Girshick R, He K, Dollár P. Focal loss for dense object detection. In: 2017 IEEE International conference on computer vision (ICCV). Venice, Italy: IEEE; 2017, p. 2999–3007. <http://dx.doi.org/10.1109/ICCV.2017.324>.
- [84] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: 2016 IEEE conference on computer vision and pattern recognition (CVPR). Las Vegas, USA: IEEE; 2016, p. 770–8. <http://dx.doi.org/10.1109/CVPR.2016.90>.
- [85] Lin T-Y, Dollár P, Girshick R, He K, Hariharan B, Belongie S. Feature pyramid networks for object detection. In: 2017 IEEE conference on computer vision and pattern recognition (CVPR). Honolulu, USA: IEEE; 2017, p. 936–44. <http://dx.doi.org/10.1109/CVPR.2017.106>.
- [86] Liu W, Anguelov D, Erhan D, Szegedy C, Reed S, Fu C-Y, Berg AC. SSD: Single shot MultiBox detector. In: Leibe B, Matas J, Sebe N, Welling M, editors. Computer vision – ECCV 2016. Cham, Switzerland: Springer; 2016, p. 21–37. http://dx.doi.org/10.1007/978-3-319-46448-0_2.
- [87] Sadykova D, Pernebayeva D, Bagheri M, James A. IN-YOLO: Real-time detection of outdoor high voltage insulators using UAV imaging. *IEEE Trans Power Deliv* 2020;35(3):1599–601. <http://dx.doi.org/10.1109/TPWRD.2019.2944741>.
- [88] Stefenon SF, Corso MP, Nied A, Perez FL, Yow K-C, Gonzalez GV, Leithardt VRQ. Classification of insulators using neural network based on computer vision. *IET Gener, Transm Distrib* 2021;16(6):1096–107. <http://dx.doi.org/10.1049/gtd2.12353>.
- [89] Liu C, Wu Y, Liu J, Han J. MTI-YOLO: A light-weight and real-time deep neural network for insulator detection in complex aerial images. *Energies* 2021;14(5):1426. <http://dx.doi.org/10.3390/en14051426>.
- [90] Chen Z, Xiao Y, Zhou Y, Li Z, Liu Y. Insulator recognition method for distribution network overhead transmission lines based on modified YOLOv3. In: 2020 Chinese automation congress (CAC). Shanghai, China: IEEE; 2020, p. 2815–20. <http://dx.doi.org/10.1109/CAC51589.2020.9327352>.
- [91] Liu M, Wang X, Zhou A, Fu X, Ma Y, Piao C. UAV-YOLO: Small object detection on unmanned aerial vehicle perspective. *Sensors* 2020;20(8):2238. <http://dx.doi.org/10.3390/s20082238>.
- [92] Chen H, He Z, Shi B, Zhong T. Research on recognition method of electrical components based on YOLO V3. *IEEE Access* 2019;7:157818–29. <http://dx.doi.org/10.1109/ACCESS.2019.2950053>.
- [93] Lewis D, Kulkarni P. Insulator defect detection. 2021. <http://dx.doi.org/10.21227/vkdw-x769>.
- [94] Rahman MA, Wang Y. Optimizing intersection-over-union in deep neural networks for image segmentation. In: International symposium on visual computing (ISVC). Advances in visual computing, Cham, Switzerland: Springer; 2016, p. 234–44. http://dx.doi.org/10.1007/978-3-319-50835-1_22.
- [95] Ferreira JC, Patino CM. What does the p-value really mean? *J Brasileiro de Pneumol* 2015;41(5):485. <http://dx.doi.org/10.1590/S1806-37132015000000215>.
- [96] Royston JP. An extension of Shapiro and Wilk's W test for normality to large samples. *J R Stat Soc Ser C Appl Stat* 1982;31(2):115–24. <http://dx.doi.org/10.2307/2347973>.
- [97] Li C, Li L, Jiang H, Weng K, Geng Y, Li L, Ke Z, Li Q, Cheng M, Nie W, Li Y, Zhang B, Liang Y, Zhou L, Xu X, Chu X, Wei X, Wei X. YOLOv6: A single-stage object detection framework for industrial applications. 2022. <http://dx.doi.org/10.48550/ARXIV.2209.02976>.
- [98] Wang C-Y, Bochkovskiy A, Liao H-YM. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. 2022. <http://dx.doi.org/10.48550/ARXIV.2207.02696>.